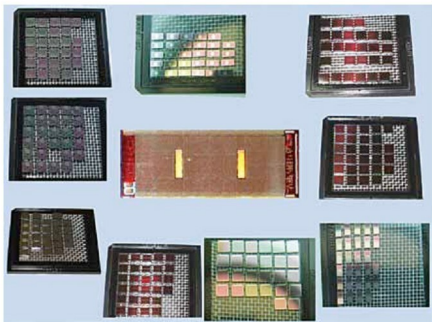


Challenges Before 2.5D/3D Technology

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Tezzaron 3D IC devices (Image courtesy: www.electroiq.com)

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A wealth of opportunities for semiconductor manufacturing are becoming available in the 'more than Moore' 2.5D/3D world. Yole Développement of France expects the shipment volume of 3D IC wafers to reach 10 million units in 2012. The semiconductor ecosystem, comprising design-houses, fabs and foundries, assembly and test providers, EDA developers and research institutes, needs to go through a learning curve as it is now necessary to have an ecosystem optimised for converting such technologies into high-volume manufacturing.

At the SEMICON Singapore 2012 event, Dr Ho-Ming Tong, GM and chief R&D officer of the Advanced

Semiconductor Engineering Inc. (ASE) Group, the world's largest provider of independent semiconductor manufacturing services in assembly and test, commented: "Despite progress in 3D-IC development over the years, challenges remain in the areas of cost control, design, mass production and testing in the lead-up to commercialisation. Given the readiness of silicon interposer-based 2.5D IC technology to move to the next stage, its deployment will expedite migration from the 40nm node to 28nm. With computing and smart devices fuelling the growth of the market, commercialisation of 2.5D and 3D ICs may take place in 2013."

To address the challenges of taking 2.5D/3D systems to high-volume manufacturing, collaborative partnerships are very critical.

"IME has established strategic partnerships with leading players in the semiconductor ecosystem to co-develop cost-effective 3D-IC integration and process technologies in Singapore. Our new 300mm advanced packaging facility will provide deeper and broader capabilities to our partners to overcome the challenges in 3D IC," informed Prof. Dim-Lee Kwong, executive director of the Institute of Microelectronics (IME)—a research institute of the Science and Engineering Research Council (SERC), Agency for Science, Technology and Research (A*STAR).

James Amano, director, International Standards, SEMI, USA, announced during SEMICON West 2012 that the SEMI 3DS-IC Committee, formed in late 2010, had approved its first standard for publication. Pending successful procedural review, the document will be published as "SEMI 3D1, Terminology for Through-Silicon Via (TSV) Geometrical Metrology."

SEMI 3D1 will provide a starting point for standardisation of geometrical metrology for selected dimensions of TSVs. The Inspection & Metrology Task Force recognised the need for such a standard because although different technologies can measure various geometrical parameters of an individual TSV or an array of TSVs—such as the pitch, top diameter, top area, depth, taper (or sidewall angle), bottom area and bottom diameter—it is currently difficult to compare results from the various measurement technologies as parameters are often described by similar names but actually represent different aspects of the TSV geometry. SEMI 3D1 is an important